

Exploring NeuroDesign in Cultivating Empathy and Creativity in Primary Education in China

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ABSTRACT

This study explores the integration of a NeuroDesign approach with an Empathy Design Thinking (EDT) curriculum to enhance creativity and empathy among primary school students in China. Utilizing functional near-infrared spectroscopy (fNIRS), the research examines the neuroscientific enhancement of educational practices. Quantitative analysis indicated trends toward improved cognitive performance, although these were not statistically significant. In contrast, qualitative data from classroom observations and interviews demonstrated notable improvements in students' empathy and creative abilities. These findings underscore the potential of NeuroDesign to significantly enrich primary education by cultivating key 21st-century skills, setting a robust groundwork for further exploration in this emergent field.

CCS CONCEPTS

- Human-centered computing - Human computer interaction (HCI)
- HCI theory, concepts and models

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KEYWORDS

NeuroDesign, Empathy Design Thinking (EDT), Primary Education, Creativity, fNIRS Technology

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1 INTRODUCTION

China's education system is undergoing transformative reforms, driven by the "Innovation and Entrepreneurship" and "Double Reduction" policies. These reforms aim to cultivate essential 21st-century skills such as creativity, critical thinking, and empathy in students [1-4]. In response, an Empathy Design Thinking (EDT) curriculum was developed specifically for primary education [5]. This program integrates theoretical concepts with practical applications to foster innovative and empathetic skills in students.

Traditionally, Chinese education has emphasized rote learning, which often underplays critical and creative thinking. The EDT curriculum challenges this norm by promoting skills necessary for innovation. This study evaluates how the EDT curriculum, integrated with NeuroDesign principles, enhances empathy and creativity among primary school students, reflecting a significant shift in educational strategies.

Our primary research question is: How does the EDT curriculum, informed by recent educational reforms and enhanced with

NeuroDesign, impact students' empathy and creativity? Addressing this question helps assess the curriculum's role as a transformative force in educational reform, particularly in developing key skills for modern challenges.

2 RELATED WORKS

The EDT curriculum represents a significant shift in primary education paradigms, rooted in holistic and comprehensive design thinking principles with a focus on empathy [6,7]. Integrating empathy as its core, this curriculum enhances both emotional intelligence and creativity among students, enriching the intellectual and emotional dimensions of education to equip students for the complexities of a globalized society [8-10].

Central to the EDT curriculum is the human-centered design (HCD) philosophy, which departs substantially from traditional educational frameworks [11]. HCD emphasizes understanding and addressing the unique needs, experiences, and aspirations of students, fostering a learning environment that prioritizes student engagement and agency—crucial elements for effective education. Aligned with modern educational methods focusing on student experiences over traditional, teacher-led instruction, HCD advocates for a dynamic and interactive learning process, enabling students to take ownership of their educational journeys [12,13].

The curriculum's innovative edge is further enhanced by the integration of a NeuroDesign approach, merging cognitive neuroscience with EDT principles to explore the neural foundations of learning and creativity [14-16]. This transdisciplinary approach optimizes teaching methods by aligning educational experiences with natural learning processes. NeuroDesign deepens our understanding of how students process and engage with new information, facilitating the development of diverse instructional strategies that accommodate various learning styles.

The commitment to empirical validation underpins the EDT curriculum, ensuring it complements and advances China's educational reforms. This research-driven focus is essential for assessing the curriculum's impact on student empathy and creativity, contributing significantly to educational reform and innovation [17-19]. By generating evidence-based outcomes, the curriculum seeks to influence educational practice and policy, ensuring its theoretical advancements are practically applicable and aligned with current educational policy goals.

3 METHODOLOGY

3.1 Research Context

The research was conducted at the China Soong Ching Ling Science and Culture Centre for Young People's Innovation Space. This 5,080 square meter facility, themed around 'Creation and Play,' is designed to foster creativity and innovation through zones dedicated to experience, exploration, and challenges.

The EDT course, centered on space exploration, integrates STEAM disciplines—Science, Technology, Engineering, Arts, and Mathematics. Over a 12-week period, it aims to enhance creativity and practical skills among primary school students. 1) Empathy Phase: Develops empathetic skills essential for human-centered design, focusing on activities that enhance students' emotional

connections with others. 2) Design Phase: Engages students in brainstorming and prototype building within space exploration contexts, enhancing their creative and design thinking skills. 3) Expression Phase: Concentrates on refining and presenting projects, improving presentation skills and the ability to exchange constructive feedback.

Throughout the course, students collaborate in teams, simulating real-world teamwork to foster a collaborative spirit and equip them with essential skills for future challenges. The curriculum combines theoretical knowledge with practical workshops, providing a dynamic and immersive learning experience that prepares students for complex situations. This approach not only achieves key educational outcomes but also lays the groundwork for further research into the integration of innovative educational practices.

3.2 Experimental Design

This study utilizes a within-subjects experimental design to evaluate the effectiveness of the EDT curriculum in enhancing cognitive abilities among primary school students. The research focuses on assessing cognitive performance variations before and after the implementation of the curriculum. Two executive function tasks, the flanker task and the switch task, were employed due to their robustness in cognitive evaluation [20,21]. Additionally, functional near-infrared spectroscopy (fNIRS) was used to provide neuroscientific insights into the activation of the frontal and parietal lobes during these tasks. This aligns the research with contemporary understanding of cognitive processes and allows for a more detailed analysis of how the EDT curriculum impacts cognitive abilities [22,23]. By combining behavioral tasks with neuroscientific measurements, the study offers a comprehensive evaluation of the EDT curriculum's potential to enhance cognitive development in young learners. See Figure 1 for an impression.

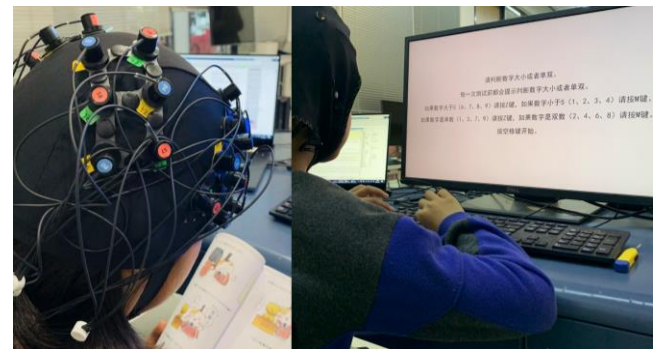


Figure 1: The fNIRS study setup.

The flanker task is employed to measure attentional inhibition, presenting congruent ("<<<<<" and ">>>>>") and incongruent ("<<<<<" and ">>>>>") stimuli in a trial-based format. Participants' responses to the direction of the central arrow assess their ability to disregard extraneous information, crucial for evaluating cognitive control. This task was chosen for its direct relevance to assessing

the cognitive skills emphasized by the EDT curriculum, particularly empathy and creative thinking [24].

The switch task builds on the flanker paradigm by adding pre-stimulus cues that signal the task condition (odd/even or low/high), requiring children to categorize numbers (1-9, excluding 5) accordingly. Featuring 'switch' and 'repeat' trials, it assesses cognitive flexibility, an essential skill that the curriculum aims to enhance, reflecting its impact on fostering empathy and creativity.

Throughout the course, detailed observations were made to capture student interactions, behaviors, and engagement levels, providing critical insights into how cognitive skills are applied within actual classroom settings and illuminating group dynamics and individual strategies. Post-course, semi-structured interviews with students, teachers, and parents were conducted to gather subjective experiences and perceptions of the curriculum's impact on cognitive, emotional, and social dimensions. These interviews were particularly focused on eliciting reflections on changes in empathy, creativity, and adaptability, enriching the understanding of the curriculum's effectiveness.

fNIRS technology was strategically utilized to track brain activity during task performance, with sensors placed on specific brain regions involved in cognitive processes. Data from fNIRS were carefully analyzed to compare neural activation patterns before and after exposure to the curriculum, providing a deeper understanding of the neurophysiological impacts of the educational interventions. The combination of cognitive tasks and fNIRS, augmented by qualitative insights, establishes a robust framework for evaluating the cognitive and socio-emotional aspects of the curriculum. This comprehensive approach is critical for demonstrating how effectively the curriculum fosters skills vital in contemporary educational settings, providing substantial evidence of its potential to transform primary education.

3.3 Participants

This study involved 30 children—14 boys and 16 girls, aged 8 to 11 years (mean age 8.49, SD = 1.12)—recruited through online platforms and community outreach efforts, including flyer distribution. Participants had normal or corrected-to-normal vision and no history of learning disabilities or neurological disorders, ensuring suitability for the demanding tasks of the study. Due to COVID-19 constraints, 20 children (8 boys and 12 girls) completed the course. This demographic and health screening was crucial to uphold the validity and reliability of the study, given the nature of the assessment methodologies employed. In line with ethical research practices, participation was strictly voluntary, accompanied by rigorous confidentiality protocols to protect participants' privacy and personal data.

4 RESULTS

The study utilized fNIRS to measure brain activity during the flanker and switch tasks, designed to assess attention inhibition and cognitive flexibility respectively (see Figure 2). Resting state analysis revealed no significant pre- and post-test differences across retained channels, suggesting potential impacts in areas excluded due to signal quality issues.

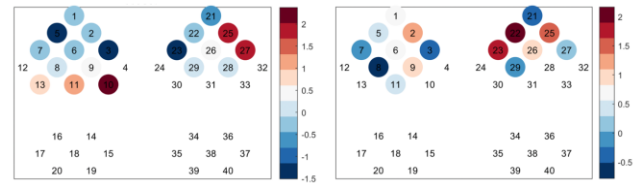


Figure 2: Task-state data analysis. Pre- and post-test comparison. Left: flanker task; right: switch task.

In the flanker task, pre-test comparisons showed no significant differences between congruent and incongruent stimuli. However, post-test analysis indicated a subtle reduction in the activation difference for attention inhibition in channel 10, although this change did not meet the threshold for statistical significance. Notably, correlations were observed between children's self-rated empathy, especially in emotional regulation, and activation differences in channels 1, 10, 11, and 28. Emotional contagion was notably related to channel 25, and parent-rated empathy dimensions correlated with channel 2. While these findings were not statistically significant after correction, they suggest an underlying link between neural activation patterns and emotional regulation. Similarly, the switch task demonstrated a decrease in activation differences related to cognitive flexibility in channel 22, yet this too did not reach statistical significance. Preliminary correlations suggested a relationship between children's self-rated empathy in perspective-taking and changes in cognitive flexibility, indicating a trend worthy of deeper investigation.

To complement the quantitative assessments, qualitative insights were gathered through observation notes and semi-structured interviews. Observations during the course revealed increased empathetic engagement among students. One teacher noted substantial growth in students' capacity to understand diverse perspectives, especially in collaborative settings. Interviews corroborated this enhanced empathy, with students, teachers, and parents noting significant improvements in emotional intelligence. A parent remarked on a profound change in their child's empathy, stating the child "sees the world more vividly." Teachers also observed a heightened ability in students to engage creatively with tasks, and students reported an improved capacity to generate ideas and connect with peers.

The combined findings from the fNIRS data and qualitative feedback, while not yielding statistically significant results, indicate promising trends in the enhancement of empathy and creativity due to the curriculum. These observations underscore the curriculum's potential efficacy in fostering essential skills for modern educational challenges.

5 DISCUSSION

5.1 Interpreting the Findings and Neuroscientific Integration

NeuroDesign, merging neuroscience with design principles, represents a novel approach in educational research. Although the

fNIRS data did not yield statistically significant differences, the observed trends suggest a subtle yet meaningful influence of the NeuroDesign curriculum on brain functions related to empathy and creativity. These preliminary findings indicate that the curriculum may affect neural processes, even if these changes are not immediately detectable at a statistically significant level. The subtle variations in cognitive flexibility and attention inhibition underscore the potential for neuroscientific methods to be integrated into educational practices, enhancing the overall learning experience. Despite operational challenges and a small sample size, the trends observed in this study highlight the value of incorporating NeuroDesign principles. This study's approach aligns with contemporary neuroscientific understanding, offering a promising avenue for future educational innovations. Addressing these challenges in future research can refine the methodology and strengthen the evidence base, potentially leading to more significant and detectable impacts on students' cognitive and emotional development.

5.2 Qualitative Insights and Enhancing Empathy and Creativity

The qualitative data collected through observations and interviews provide compelling evidence of the curriculum's impact [25]. Teachers reported significant improvements in students' empathetic interactions, noting increased awareness of peers' emotions and perspectives. Parents shared similar observations, seeing greater sensitivity and understanding in their children. Students themselves expressed feeling more connected to their classmates and more confident in their creative abilities [26]. These qualitative insights demonstrate the curriculum's effectiveness in fostering essential social and emotional skills [27]. Enhancements in empathy and creativity observed in this study are critical for personal and academic success, underscoring the real-world applicability and benefits of incorporating NeuroDesign principles into educational curricula. These findings highlight the importance of integrating both empathy and creativity into educational practices, aligning with contemporary educational reforms and emphasizing the need for further research to validate and expand these initial positive outcomes.

5.3 Aligning with Educational Paradigms and Policy Implications

This research aligns with China's evolving educational paradigms, emphasizing holistic development and the cultivation of 21st-century skills over traditional rote learning methods [28]. The EDT curriculum's focus on empathy and creativity supports these policy shifts, demonstrating practical relevance and effectiveness. By fostering these core competencies, the curriculum helps students develop the skills needed to navigate an increasingly complex and interconnected world. The positive outcomes observed in this study suggest that such innovative educational approaches can significantly contribute to national educational goals and inform policy formulations aimed at improving holistic student development. These findings underscore the importance of

integrating empathy and creativity into educational practices to meet contemporary educational challenges and policy objectives.

5.4 Challenges, Limitations, and Future Directions

Implementing NeuroDesign in education presents several challenges, as highlighted by the limitations encountered in this study. The use of fNIRS technology, although innovative, faced operational challenges affecting data quality and interpretation [29–31]. The relatively small and demographically narrow sample size limits the generalizability of the findings, emphasizing the need for larger and more diverse participant groups in future research. Additionally, integrating NeuroDesign into the curriculum proved challenging, requiring ongoing refinement to balance theoretical concepts with practical teaching strategies. These limitations necessitate cautious interpretation of the results and reflect the complexities of embedding advanced neuroscientific methods into educational contexts. Future research should address these challenges by expanding sample sizes, improving data collection methods, and conducting longitudinal studies to assess the sustained impact of the NeuroDesign curriculum on student development.

6 CONCLUSIONS

This study explores the use of EDT, enhanced by NeuroDesign, in primary education to foster empathy and creativity. Qualitative data indicate that the EDT curriculum positively impacts these skills, even with challenges using fNIRS technology. These results emphasize the potential of integrating neuroscientific insights into educational practices to enhance student learning.

The curriculum's focus on empathy and creativity aligns with China's educational policy shifts, which prioritize holistic development over rote learning. By fostering 21st-century skills, EDT helps students navigate a complex, interconnected world. Improvements in empathy and creativity highlight the promise of applying neuroscientific approaches in education.

Despite limitations such as a small sample size and operational challenges with fNIRS, the study shows the potential of EDT informed by NeuroDesign. Future research should involve a more diverse participant pool and refine data collection methods to validate its effectiveness.

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